Doubling, truncation and fusion joins for the $\mathrm{PSL}_3(4)$ catalogue

L100 finite source-join component

We use the fixed Halle preprint of Gottschalk–Leemans [1]; the later journal edition remains uncollated. Its labels $2.1, \ldots, 5.4$ are joined to the accepted 29 literal coset representatives by the previous source-join component. The complete subgroup inputs, broad primitive enumeration, full-automorphism fusion and palette calculations retain their exact previously stated premises. In particular local two-transitivity was tested, not imposed as an enumeration filter.

Theorem 1. *The four printed doubling links are valid:*

$$2.7 \longrightarrow 2.6, \quad 4.1 \longrightarrow 4.8, \quad 5.2 \longrightarrow 5.3, \quad 5.1 \longrightarrow 5.4.$$

Of the 24 printed truncation links, 23 have complete literal incidence maps. The remaining assertion, on physical page 9, that 3.6 *is a truncation of* 5.3 *is false by type cardinalities; replacing* 5.3 *by* 5.2 *gives a verified truncation. All 29 printed parenthetical fusion multiplicities agree with the complete acquired 145-family partition.*

The truncation error concerns an ancillary link. It does not refute the principal classification. This component does not identify the abstract names of parabolics, kernels, automorphism or correlation groups.

1 A concrete doubling argument

Lemma 1. *Let Γ be a flag-transitive geometry with point type X and another type E. Suppose the X-shadows of the objects of E are distinct two-element sets. Assume that, for every object z of any other type,*

$$e * z \quad \Longleftrightarrow \quad \sigma_X(e) \subseteq \sigma_X(z). \tag{1}$$

Replace X, E by copies X^+, X^- of X. Declare $x^+ * y^-$ when $\{x, y\} = \sigma_X(e)$ for some $e \in E$; declare either copy of x incident with z exactly when $x * z$ in Γ; leave all other incidences unchanged. The resulting incidence system is a geometry, and the induced specified group action is flag-transitive.

Proof. A flag using both copies gives distinct points x, y and a unique object e with shadow $\{x, y\}$. By (1), replacing y^- by e gives an old flag through x, e and the unchanged objects. Extend it to an old chamber. Its associated new chamber extends the original flag. A flag using only X^+ extends from the corresponding old flag through x; the other point of its chamber's E-object supplies the minus copy. With only X^- interchange these roles. With neither, extend the unchanged old flag first. Thus every new flag extends.

The chamber bijection sends (x, e, Z) to (x^+, y^-, Z). Here y is the other point of e, and the bijection is equivariant. Consequently old chamber transitivity implies new chamber transitivity; in a geometry this implies flag transitivity by extending flags to chambers. The proof works in rank two as well. □

Condition (1) also implies the particular (I_{12}) condition printed on page 6. For an object e and an object z of another type, their point-shadow intersection is empty, a singleton $\{x\}$, or the full shadow of e. In the singleton case the flag $\{x\}$ is incident with both and has the required point shadow. In the full case (1) gives $e * z$, so the flag $\{e\}$ works. Incidence here is reflexive, as in the source's coset definition.

The source's quoted Corollary 2.1 is explicitly a rank-at-least-three statement attributed to its reference [40]. Its use at 2.7 $\to$ 2.6 therefore requires a qualification. The direct rank-two argument above supplies that case: 2.7 is the edge-incidence system of a graph, and the construction is its bipartite double. We do not extend the scope of the quoted corollary by assertion.

For $4.1, 5.1, 5.2$, the accepted projective model supplies ordinary points and unordered point pairs. Incidence with a hyperoval means containment, and incidence with the relevant line means disjointness, for both the point and pair types. Thus (1) also has this immediate projective explanation. Independently, the finite verifier checks every pair shadow, its uniqueness, and every instance of (1) in the actual coset models. No source point or line label is assumed to equal a coset label.

2 Exact identifications and all truncations

All new maps use the same positive certificate interface. If H_i is an actual target parabolic, the certificate gives a derived source object x_i fixed by its images under an explicitly checked group automorphism α. On target right cosets the map is

$$H_i g \longmapsto x_i^{\alpha(g)}.$$

The derived geometry is either a doubled model from the lemma or an induced system on a specified subset of source types. The original coset actions are checked against the saved generator permutations; right multiplication, not a guessed handedness, defines all object images. Allowed automorphisms are literal conjugations by acquired point normalizers and the acquired coordinate duality. Their exact generator images and full group bijections are retained.

For every map the verifier checks all coset members, all typewise bijections, every incidence and nonincidence row, and every generator action on every object. Thus the target's full geometry and residue properties transport, including its actual local two-transitivity flags. This argument does not infer a map from matching group orders, diagrams or object counts. The four doubling maps require respectively $40320, 80640, 100800, 100800$ coset-member checks.

The complete positive truncation ledger is as follows. Eight links were already accepted in the projective-construction component and are acquired unchanged; sixteen additional maps are checked here.

Target	Verified parent geometries
2.2	5.1, 5.2, 4.10, 3.1, 3.4, 3.6
2.4	4.11, 3.5
2.5	3.3
3.1	5.1, 5.2, 4.1
3.2	4.3, 4.4
3.6	5.1, **5.2**, 4.1, 4.10
4.1	5.1, 5.2
4.8	5.3, 5.4
4.9	5.2
4.10	5.1

The bold entry repairs the printed 5.3. Indeed, a truncation retains the complete object sets of selected types. The type sizes of 3.6 are $56, 210, 56$, whereas those of 5.3 are $21, 21, 21, 56, 56$; these are also visible directly in the native diagrams on pages 9 and 13. No truncation of 5.3

can introduce a 210-object type. The replacement 5.2 is established positively by its saved map. It is also consistent with the projective constructions: 5.1 and 5.2 differ only in a line–hyperoval relation, whereas this truncation keeps the two hyperoval types and the point-pair type.

3 Fusion multiplicities and evidence boundary

The acquired complete fusion records contain 145 distinct inner-conjugacy families with type permutations allowed, partitioned by the actual full $\mathrm{Aut}(G)$ action. The rank totals are $21, 25, 63, 36$ for ranks $2, 3, 4, 5$. Each retained family occurs once. The 29 representative identities agree exactly with the independently acquired bare catalogue. Grouping the saved membership records by these exact identities gives the following source-order multiplicities:

Rank	Multiplicities in increasing source-label order
2	1,6,3,3,6,1,1
3	6,3,3,3,3,6,1
4	6,3,3,6,6,6,3,6,6,12,6
5	12,6,6,12

These agree with all parenthetical numbers on pages 7–13. The families are counted modulo G-conjugacy and type permutations, then fused under $\mathrm{Aut}(G)$, not extra bare geometries or correlation automorphisms. The projection parser accepts only the complete restricted membership-record syntax; it checks all 145 distinct member identities and all 29 representative joins. The completeness and inequivalence of those records are acquired from the accepted enumeration and fusion proofs, not re-proved by counting their rows.

The first truncation pass saved ten complete positive maps before rejecting the impossible printed link. That failed terminal receipt is preserved. A changed pass handled only the six outstanding positive maps, including the explicit replacement. Default replay verifies all twenty new maps directly and does not repeat candidate-map discovery, an ambient census or a graph-isomorphism search. Small-model controls check the doubling construction and chamber correspondence; separate damaged maps and record controls enforce the exact interfaces. Abstract structural names, ordered-butterfly and oval constructions, and the final whole-source synthesis remain separate. There is no new measurement, Lean, question or priority claim.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* $\mathrm{PSL}(3, 4)$, Halle Reports on Algebra and Geometry 99–30 (1999), fixed 22-page preprint.